Auteur: Siebe Mees
Studierichting: Informatica
Oefeningenreeks 3H nr. 11.3

$r^2 = 9\sin 2\theta \Leftrightarrow r = \sqrt{9\sin 2\theta}$

1. Domein

   $p = \frac{2\pi}{2} = \pi$

   $\Rightarrow$ we onderzoeken op het beperkt domein $[0, \pi]$

   Binnen dit interval geldt: (Omdat in $\mathbb{R}$ de wortel van een negatief getal niet bestaat)

   $\sin 2\theta \geq 0 \iff \theta \in \left[0, \frac{\pi}{2}\right]$

   $\Rightarrow \operatorname{dom} r = \left[0, \frac{\pi}{2}\right]$

2. Tekenonderzoek van $r(\theta)$

   $r^2 = 9\sin 2\theta$

   $\sin 2\theta = 0$

   $2\theta = k\pi$

   $\theta = \frac{k\pi}{2}$

   $\Rightarrow r^{(-1)}(0) = \left\{0, \frac{\pi}{2}\right\}$

θ	0		$\frac{\pi}{2}$
$r(\theta)$	0	+	0

3. Tekenonderzoek van $r'(\theta)$

   $r = \sqrt{9\sin 2\theta}$

   $r = 3\sqrt{\sin 2\theta}$

   $r(\theta) = 3(\sin 2\theta)^{\frac{1}{2}}$

$$r'(\theta) = 3 \cdot \frac{1}{2}(\sin 2\theta)^{-\frac{1}{2}} \cdot 2\cos 2\theta$$

$$r'(\theta) = \frac{3}{\sqrt{\sin 2\theta}} \cos 2\theta$$

$$\cos 2\theta = 0$$

$$2\theta = \frac{\pi}{2} + k\pi$$

$$\theta = \frac{\pi}{4} + \frac{k\pi}{2}$$

$$\Rightarrow r'^{(-1)}(0) = \left\{\frac{\pi}{4}\right\}$$

θ		$\frac{\pi}{4}$	
$r'(\theta)$	$+$	0	$-$

Maximum:

$$r\left(\frac{\pi}{4}\right) = \sqrt{9 \sin\left(2 \cdot \frac{\pi}{4}\right)}$$

$$r\left(\frac{\pi}{4}\right) = \sqrt{9 \sin \frac{\pi}{2}}$$

$$r\left(\frac{\pi}{4}\right) = \sqrt{9}$$

$$\Rightarrow \max\ r\left(\frac{\pi}{4}\right) = 3$$

4. Schets

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$\Rightarrow$ Samenvattende tabel

θ	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$
$r(\theta)$	0	$+$	0
$r'(\theta)$	$+$	0	$-$
$r(\theta)$	$\nearrow$	$M(3)$	$\searrow$

References